
Learning Spatial Unit-Norm Latents with Riemannian Flow Matching

Mohammadreza Tavasoli Naeini

Department of Electrical and Computer Engineering, University of Toronto
mohammadreza.tavasoliniaeini@mail.utoronto.ca

Abstract

Latent generative models often use a Gaussian prior, but autoencoders may instead produce nearly constant-norm features; normalized Transformer encoders are one example. We study a simple rule: the prior should match the latent support learned by the autoencoder. Spherical Riemannian Unit-Norm Latents (SRUL) trains spatial unit-norm tokens and models them with Riemannian Flow Matching (RFM) on the same product-of-spheres support. Its tangent-noise scale σ_{enc} is an explicit information-control knob: it changes how much image-specific information is retained without leaving the sphere. On CIFAR-10, geodesic transport improves FID from 67.44 to 63.84 over a straight chord with the same spherical autoencoder. SRUL reaches FID 49.42 in the support comparison, versus 50.70 for VAE + standard FM and 56.04 for VAE + projected RFM. An information-control sweep and a classifier-free-guidance sweep study the main controls, while PathMNIST and CelebA-64 test additional image settings.

Attestation of Teamwork

Mohammadreza Tavasoli Naeini conducted the background study, designed and implemented the models, ran and analyzed the experiments, and prepared the report, presentation, and code.

1 Introduction

Latent generative models have two parts. An autoencoder first maps an image to a latent code, and a prior then learns how to generate new codes for the decoder. Gaussian latents are common, especially in VAE-based models, but the encoder is free to learn a different geometry. Normalization can make token norms nearly constant, so information is carried mainly by direction. Transformer encoders with LayerNorm are one example, and explicit L_2 normalization can create the same structure in a convolutional encoder.

The prior should therefore be chosen after considering the latent support. When an encoder produces spherical tokens, standard linear interpolation is not well matched to them: even if both endpoints have unit norm, the straight path is a chord through the sphere’s interior. The reverse mismatch can also occur. A VAE may use both token direction and token magnitude; projecting its latents onto a sphere after training removes the magnitude information that its decoder expects.

To study this support-matching principle, we introduce Spherical Riemannian Unit-Norm Latents (SRUL). Its autoencoder learns a spatial grid of unit-norm tokens, and its RFM prior learns a probability path on the same product of spheres. The representation and prior are therefore trained for the same latent support instead of being connected by a post-hoc projection.

We study the design with two controlled tests. First, we keep the spherical autoencoder unchanged and compare a Euclidean chord with geodesic RFM transport. Second, we compare SRUL with a VAE modeled by standard FM and with the same VAE after its latents are reduced to directions
ECE 1508: Deep Generative Models (Summer 2026)

for projected RFM. The second test shows whether a spherical prior can be attached to a VAE after training without losing radial information. We use the tangent-noise scale as an information-control knob, study classifier-free guidance, and evaluate the same pipeline on CIFAR-10, PathMNIST, and CelebA-64.

2 Related Work and Positioning

Variational autoencoders learn stochastic Euclidean latents and use a KL term to regularize them toward a Gaussian prior [1]. Latent Diffusion Models then move generation from pixels to such a learned latent grid, reducing computation while preserving perceptually important information [2]. Unified Latents studies information control through encoder noise in a Euclidean generative process [6]; SRUL adapts this idea to tangent perturbations on spherical support. Sphere Encoder explicitly maps images to a sphere and trains the decoder to remain stable around encoded points, with the goal of direct sampling from an approximately uniform spherical distribution [5].

Flow Matching learns a time-dependent velocity field along a chosen probability path [3]. Standard linear Flow Matching uses straight paths in Euclidean latent space. RJF observes that normalized representation encoders often produce features with nearly constant norm and replaces this interpolation with Riemannian Flow Matching on the sphere; it also studies an additional Jacobi weighting [4]. SRUL differs in two ways. It learns the spatial spherical representation rather than assuming a pretrained representation encoder, and it uses tangent perturbation as a controllable latent bottleneck. This makes encoder–prior co-design, rather than only applying RFM to an existing representation, the central question. The final method uses the RFM formulation because it was the strongest variant in our experiments.

3 Method

3.1 Pipeline overview

Figure 1 shows the full pipeline. Representation learning produces unit-norm tokens and uses tangent perturbations to control retained information while training a robust decoder. RFM then learns their distribution using geodesic paths and tangent velocities. During sampling, spherical noise is transported with exponential-map steps and decoded into a new image.

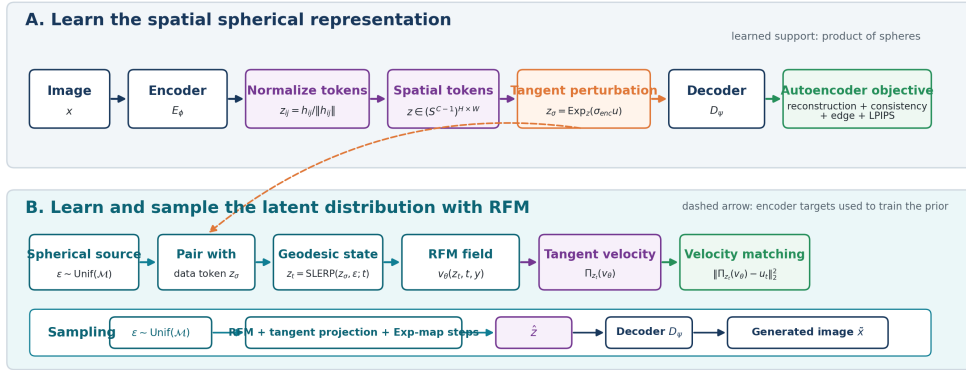


Figure 1: SRUL pipeline. The first stage learns spatial spherical tokens and uses tangent perturbations for information control. The second stage learns and samples an RFM prior on the same product-of-spheres manifold. Class labels and classifier-free guidance are optional conditional inputs.

3.2 Spatial spherical autoencoder

For an image x , the encoder produces a spatial tensor

$$h = E_\phi(x) \in \mathbb{R}^{C \times H \times W}, \quad z_{:,i,j} = \frac{h_{:,i,j}}{\|h_{:,i,j}\|_2} \in S^{C-1}. \quad (1)$$

The full latent space is $\mathcal{M} = (S^{C-1})^{HW}$. We use $C = 32$, a 4×4 token grid for 32×32 images, and an 8×8 grid for CelebA-64. The constraint is imposed during representation learning; it is not added after an unconstrained autoencoder has already decided how to use token magnitude.

The autoencoder is trained from clean and perturbed latents:

$$\mathcal{L}_{\text{AE}} = \lambda_c \|D_\psi(z) - x\|_1 + \lambda_n \|D_\psi(z_\sigma) - x\|_1 + \lambda_e \mathcal{L}_{\text{edge}} + \lambda_{\text{lat}} [1 - \cos(z, E_\phi(D_\psi(z_\sigma)))] + \lambda_p \mathcal{L}_{\text{LPIPS}}. \quad (2)$$

The two L_1 terms preserve image content from clean and perturbed codes. LPIPS compares features from a frozen pretrained network rather than pixels alone, so it rewards similar shape and texture; the edge term preserves local boundaries [10]. The latent-consistency term re-encodes the noisy reconstruction and aligns it with the clean token directions, making nearby spherical codes decode to a stable semantic neighborhood. Appendix A gives the full interpretation.

3.3 Tangent-noise information control

For each token, Gaussian noise is projected onto the tangent space,

$$\xi \sim \mathcal{N}(0, I), \quad u = \Pi_z(\xi) = \xi - \langle \xi, z \rangle z, \quad (3)$$

and the perturbed token is

$$z_\sigma = \text{Exp}_z(\sigma_{\text{enc}} u). \quad (4)$$

For a tangent vector v , the spherical exponential map is

$$\text{Exp}_z(v) = \cos(\|v\|_2) z + \sin(\|v\|_2) \frac{v}{\|v\|_2}. \quad (5)$$

Because $v \perp z$, this is a rotation and its output still has unit norm. The scalar σ_{enc} is an information-control knob: larger values perturb directions more strongly and retain less image-specific information. Appendix C gives the norm proof and sampling interpretation.

3.4 Riemannian Flow Matching prior

Standard linear Flow Matching connects a source ϵ and target z with $x_t = (1 - t)\epsilon + tz$, whose target velocity is $z - \epsilon$. RFM keeps this velocity-regression idea but replaces the straight path with a manifold geodesic. The source is uniform on the same manifold, $\epsilon \sim \text{Unif}(\mathcal{M})$. For a data latent z_σ and source sample ϵ , the tokenwise geodesic is

$$z_t = \text{SLERP}(z_\sigma, \epsilon; t) = \frac{\sin((1 - t)\Omega)}{\sin \Omega} z_\sigma + \frac{\sin(t\Omega)}{\sin \Omega} \epsilon, \quad \Omega = \arccos(z_\sigma^\top \epsilon). \quad (6)$$

SLERP is a constant-speed rotation in the two-dimensional plane of its endpoints. Therefore $\|z_t\|_2 = 1$ for every t , and the curve is the great-circle geodesic; Appendix B gives a direct proof. Let $u_t = \partial z_t / \partial t$ be the known target tangent velocity. We train

$$\mathcal{L}_{\text{RFM}} = \mathbb{E} \left[\left\| \Pi_{z_t}(v_\theta(z_t, t)) - u_t \right\|_2^2 \right]. \quad (7)$$

Thus RFM training is supervised velocity regression: a random point on a known geodesic is constructed, its exact velocity is computed, and the network is trained to predict that velocity after tangent projection. During sampling, the target endpoint is no longer known; the model starts from spherical noise and repeatedly follows its learned field,

$$z_{t-\Delta t} = \text{Exp}_{z_t}(-\Delta t \Pi_{z_t}(v_\theta)), \quad (8)$$

so every intermediate token remains on its sphere.

3.5 Conditional generation and classifier-free guidance

The field may also receive a label y , giving $v_\theta(z_t, t, y)$. During training, the label is replaced by a null label with probability 0.1. During sampling,

$$v_{\text{CFG}} = v_\theta(z_t, t, \emptyset) + s [v_\theta(z_t, t, y) - v_\theta(z_t, t, \emptyset)]. \quad (9)$$

Bayes rule motivates separating an unconditional component from a label-specific component in classifier guidance. CFG learns both components in one network by randomly dropping labels, then amplifies their difference during sampling [11]. The guided field is projected before every exponential-map update, so conditioning changes sampling strength but not the manifold. Larger s usually pushes samples toward more class-typical regions, which can improve FID and precision while reducing coverage and recall; Appendix D explains this relation carefully.

4 Experiments

Setup

CIFAR-10 is used for controlled geometry and support-matching tests [7]. PathMNIST evaluates transfer to histopathology texture [8], and CelebA-64 evaluates a larger 64×64 setting [9]. We use AdamW in both training stages and 100 integration steps for final sampling. We report FID, KID, feature precision and recall, and token norm.

Does the encoder–prior geometry match?

We test the same design principle at two levels. The first experiment asks whether transport remains on a spherical support that is already defined by the encoder. The second asks whether a spherical direction-only prior can be attached after a Euclidean autoencoder has learned a different support.

Transport on spherical support. The path comparison uses the same spherical encoder, decoder, source distribution, latent size, prior capacity, and training budget. Only the path changes. The baseline uses

$$z_t^{\text{chord}} = (1 - t)z_\sigma + t\epsilon. \quad (10)$$

Although both endpoints have unit norm, the chord enters the sphere. For nearly orthogonal endpoints, its midpoint norm is approximately $1/\sqrt{2} = 0.707$; the measured minimum mean norm is 0.705. SRUL follows SLERP and remains at norm 1.

Same spherical endpoints, different transport path

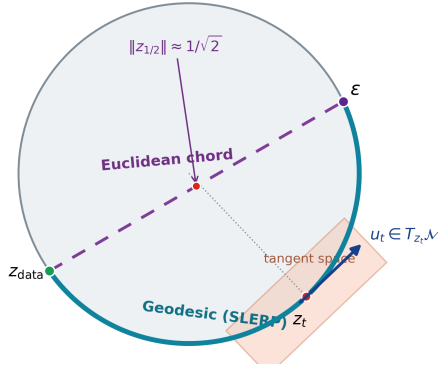


Figure 2: With the same spherical encoder, the chord visits interior states while SRUL follows the geodesic and keeps its velocity in the tangent space.

Table 1: CIFAR-10 path comparison with the same spherical autoencoder.

Method	FID↓	KID↓	Precision↑	Recall↑	Min. norm↑
Chord baseline	67.44	0.0740	0.840	0.085	0.705
SRUL	63.84	0.0667	0.857	0.091	1.000

The geodesic improves all four image metrics and removes the radial excursion. This isolates the benefit of matching the transport path to an already-spherical representation.

Support matching with the same VAE. A VAE is a practical Euclidean latent model because KL regularization encourages a Gaussian-like latent, while its decoder still receives the complete latent vector. We keep one trained VAE encoder and decoder unchanged and vary only the prior. Standard FM models the full token z . Projected RFM keeps only its direction $u = z/\|z\|_2$ and restores a class/token mean radius before decoding. SRUL is shown in the same conditional evaluation because it learns spherical support before training its RFM prior.

Table 2: CIFAR-10 support-matching comparison (CFG $s = 2$).

Method	Latent support and prior	FID↓	Precision↑	Recall↑
SRUL	Spherical autoencoder + RFM	49.42	0.857	0.110
VAE + standard FM	Full VAE latent + linear FM	50.70	0.855	0.091
VAE + projected RFM	Directions + class/token mean radius	56.04	0.848	0.063

The VAE token radii have coefficient of variation 0.291, so magnitude is not constant. Replacing sample-specific radii with class/token means raises reconstruction FID from 16.90 to 22.46; forcing unit radius gives 113.00. Standard FM preserves the full VAE latent and reaches FID 50.70. Projected RFM removes sample-specific radius information and worsens FID to 56.04. SRUL learns spherical support before RFM and reaches FID 49.42. The result shows that the autoencoder and prior should be trained for the same support.

Information-control and guidance ablations

We train a separate conditional SRUL prior for each value of the information-control knob σ_{enc} . Larger values remove more image-specific information and worsen both reconstruction and generation; 0.05 is the best tested value. We then vary CFG while leaving the representation and RFM objective unchanged. Stronger guidance improves FID and precision, but reduces recall.

Table 3: CIFAR-10 information-control and classifier-free-guidance ablations.

σ_{enc}	Recon. FID↓	Gen. FID↓	Recall↑	CFG s	FID↓	Precision↑	Recall↑
0.05	17.53	48.71	0.090	1.0	62.44	0.854	0.083
0.15	19.56	51.61	0.090	1.5	54.83	0.865	0.071
0.30	27.85	58.16	0.073	2.0	49.39	0.884	0.059

Evaluation across image domains

The same spatial spherical representation and RFM prior are evaluated in three settings. CIFAR-10 provides controlled geometry and sampling tests, PathMNIST evaluates texture-dominated histopathology, and CelebA-64 tests a larger token grid on structured facial images.

Table 4: Selected SRUL results. Each row is evaluated against the real split of that dataset.

Dataset	Evaluation setting	Grid / selected controls	Recon. FID	Gen. FID	Precision	Recall
CIFAR-10	Natural-object geometry and sampling study	4×4 ; $\sigma_{\text{enc}} = 0.05$, $s = 2$	17.53	48.71	0.861	0.090
PathMNIST	Histopathology texture transfer	4×4 ; $\sigma_{\text{enc}} = 0.15$, $s = 2$	7.13	29.36	0.829	0.309
CelebA-64	64×64 facial-image scale-up	8×8 ; $\sigma_{\text{enc}} = 0.15$, $s = 2$	9.08	41.22	0.657	0.198

The same design transfers to PathMNIST tissue texture and scales to an 8×8 grid for CelebA-64 faces.

5 Conclusion

SRUL trains spatial unit-norm tokens and an RFM prior on the same product-of-spheres support. With the spherical autoencoder unchanged, geodesic transport improves FID from 67.44 to 63.84 over a straight chord. With the same trained VAE, standard FM on the full latent reaches 50.70, while post-hoc projection to directions before RFM worsens FID to 56.04. Its tangent-noise scale also provides explicit information control. The results support a simple conclusion: the autoencoder should learn the support that its prior is designed to model.

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A Autoencoder losses in detail

The autoencoder loss in Eq. (2) combines image-level fidelity with latent-space stability. The terms have different roles and should not be interpreted as interchangeable versions of the same error.

Pixel, edge, and perceptual losses. The clean and noisy L_1 terms compare corresponding pixels. They preserve color, brightness, and approximate spatial location, but pixel distance can penalize a small shift strongly and can also reward blurry averages. The edge loss compares horizontal and vertical image gradients, so it gives direct pressure to preserve boundaries and local structure. LPIPS instead passes the real and reconstructed images through a frozen pretrained feature network and compares normalized features at several layers [10]. A simplified form is

$$\mathcal{L}_{\text{LPIPS}}(x, \hat{x}) = \sum_{\ell} \frac{1}{H_{\ell} W_{\ell}} \sum_p \|w_{\ell} \odot (\bar{f}_{\ell}(x)_p - \bar{f}_{\ell}(\hat{x})_p)\|_2^2. \quad (11)$$

Early features respond to edges and textures, while deeper features respond to larger visual patterns. The feature network is frozen; gradients pass through it only to update the SRUL encoder and decoder.

Latent consistency. Let

$$\tilde{z} = E_{\phi}(D_{\psi}(z_{\sigma})). \quad (12)$$

The latent-consistency term is the average cosine distance between the clean token z and the re-encoded noisy reconstruction $\tilde{z}$:

$$\mathcal{L}_{\text{lat}} = 1 - \frac{1}{BHW} \sum_{b,i,j} z_{b,i,j}^{\top} \tilde{z}_{b,i,j}. \quad (13)$$

Because both tokens are unit norm, their dot product is exactly the cosine of their angle. A loss of zero means that the noisy round trip returns to the same token direction. In the implementation, the clean z is treated as a stopped-gradient target for this branch. The term therefore acts as a local denoising and cycle-consistency constraint: nearby spherical codes should decode to the same image content and re-encode near the same clean representation.

B From standard Flow Matching to spherical RFM

B.1 Standard linear Flow Matching

Let $\epsilon \sim p_0$ be a simple source and $z \sim p_1$ a target latent. Standard linear Flow Matching chooses

$$x_t = (1-t)\epsilon + tz, \quad u_t = \frac{\partial x_t}{\partial t} = z - \epsilon. \quad (14)$$

A neural field is trained by

$$\mathcal{L}_{\text{FM}} = \mathbb{E}[\|v_{\theta}(x_t, t) - u_t\|_2^2]. \quad (15)$$

At inference, one samples $x_0 \sim p_0$ and solves the ODE $\dot{x}_t = v_{\theta}(x_t, t)$. Training is called simulation-free because a full ODE trajectory is not solved for every update: one samples t , constructs x_t directly, and regresses its known velocity [3].

B.2 Why SLERP is the spherical geodesic

Let $a, b \in S^{d-1}$, let $\Omega = \arccos(a^{\top} b)$, and assume $0 < \Omega < \pi$. Define

$$q = \frac{b - \cos \Omega a}{\sin \Omega}. \quad (16)$$

Then $q \perp a$, $\|q\|_2 = 1$, and $b = \cos \Omega a + \sin \Omega q$. Substituting this expression into Eq. (6) gives

$$\gamma(t) = \cos(t\Omega)a + \sin(t\Omega)q. \quad (17)$$

This is a rotation through angle $t\Omega$ in the plane spanned by the endpoints. Since a and q are orthonormal,

$$\|\gamma(t)\|_2^2 = \cos^2(t\Omega) + \sin^2(t\Omega) = 1. \quad (18)$$

The speed is constant, $\|\dot{\gamma}(t)\|_2 = \Omega$, and

$$\ddot{\gamma}(t) = -\Omega^2 \gamma(t). \quad (19)$$

On the unit sphere, $\gamma(t)$ is the normal direction. Hence the acceleration has no tangential component, which is the geodesic equation. For non-antipodal endpoints, this great-circle arc is the unique shortest geodesic.

B.3 RFM target and loss

Differentiating SLERP gives the tokenwise target velocity

$$u_t = \frac{\Omega}{\sin \Omega} [-\cos((1-t)\Omega)a + \cos(t\Omega)b]. \quad (20)$$

Because $\|\gamma(t)\|_2 = 1$, differentiating the norm gives $\gamma(t)^\top u_t = 0$; the target is tangent. The neural network may output a radial component, so its prediction is projected as

$$\Pi_z(v) = v - (z^\top v)z. \quad (21)$$

Equation (7) then matches the projected prediction to the exact tangent velocity and averages the error over the batch and all spatial tokens.

C Exponential-map sampling

At inference there is no paired target latent. Sampling starts from $z_0 \sim \text{Unif}(\mathcal{M})$ and repeatedly queries the learned field. A direct Euler step is not manifold preserving. Even when $v \perp z$ and $\|z\|_2 = 1$,

$$\|z + \Delta t v\|_2^2 = 1 + \Delta t^2 \|v\|_2^2 > 1. \quad (22)$$

The exponential map instead performs a geodesic step:

$$z^+ = \text{Exp}_z(\Delta t v) = \cos(\Delta t \|v\|_2)z + \sin(\Delta t \|v\|_2) \frac{v}{\|v\|_2}. \quad (23)$$

Since z and $v/\|v\|_2$ are orthonormal, $\|z^+\|_2^2 = \cos^2(\cdot) + \sin^2(\cdot) = 1$. Each sampling step therefore remains on the sphere. In SRUL, the operation is applied independently to every spatial token:

1. evaluate the conditional or unconditional velocity field;
2. apply classifier-free guidance when used;
3. project the final velocity onto each token’s tangent space;
4. update every token with the exponential map;
5. decode the final token grid after the last integration step.

D Classifier-free guidance and the fidelity–coverage trade-off

Bayes rule motivates classifier guidance at the score level. For a noisy state z_t and label y ,

$$\nabla_{z_t} \log p_t(z_t | y) = \nabla_{z_t} \log p_t(z_t) + \nabla_{z_t} \log p_t(y | z_t). \quad (24)$$

Classical classifier guidance estimates the second term with a separate noisy-image classifier. Classifier-free guidance avoids this extra model by training one generator both conditionally and unconditionally: labels are dropped on a small fraction of training examples [11]. At inference it extrapolates from the unconditional field toward the conditional field. In our velocity parameterization this is Eq. (9):

$$v_{\text{CFG}} = v_\emptyset + s(v_y - v_\emptyset). \quad (25)$$

For $s = 1$, this equals the ordinary conditional field. For $s > 1$, the difference $v_y - v_\emptyset$ is amplified. The Bayes identity is exact for score fields; applying the same linear combination to RFM velocity fields is the classifier-free guidance rule used in our implementation, not a new Bayes identity for velocities.

Stronger guidance tends to move samples toward regions that are highly characteristic of the requested class. This can increase class fidelity and feature precision and can improve FID, but it also concentrates probability on fewer modes, so recall may decrease. This is an empirical trade-off rather than a theorem. On CIFAR-10, increasing s from 1 to 2 changes FID from 62.44 to 49.39 and recall from 0.083 to 0.059.